DCS Greece

Lockon Greece EWRS

Player & Mission Designer Guide • Version 11.8

Early Warning Radar System

DCS World Mission Scripting • Requires MIST

1. Overview

The Lockon Greece EWRS (Early Warning Radar System) is a Lua script for DCS World that provides players with a GCI-style air picture via the in-game F10 radio menu and automatic text reports. It collects radar contacts from all friendly ground EWR stations, SAM radars, AWACS, ships, and other configured sensor types, then broadcasts bearing, range, altitude, aspect, and optional speed information to subscribed players.

1.1 Key Features

- Automatic GCI-style contact reports on a configurable refresh timer.
- On-demand Bogey Dope — instant enemy picture without waiting for the next refresh.
- Friendly Picture — shows positions of friendly aircraft in your vicinity.
- Training Mode — instead of the enemy air picture, shows which enemy radars are currently tracking you and from what bearing/range. Ideal for SEAD and defensive flying practice.
- Per-player settings: range filter, max contacts, units (imperial/metric), bearing reference (ownship or bullseye), helicopter inclusion, and speed reporting.
- Separate Blue and Red instances — each coalition sees only its own picture.
- Automatic menu cleanup on death, ejection, crash, and slot change.
- Dynamic radar cache — picks up units spawned after mission start.

1.2 Dependencies

- MIST (Mission Scripting Tools) — must be loaded before EWRS.
- DCS World with Scripting Engine enabled in the mission options.

⚠ EWRS will display an error message and abort if MIST is not loaded first.

2. Setup for Mission Designers

2.1 Script Load Order

In your mission ONCE triggers, add DO SCRIPT FILE actions in this order:

Order	Script
1	MIST.lua (required foundation library)
2	DCS_Greece_-_Lockon_Greece_EWRS_v11_8.lua
3 (optional)	IADS_v5_linked_ewr_VTG.lua (if using the SAM zone manager)

2.2 Verifying the Script Is Running

When the mission loads, a confirmation message will briefly appear on-screen for all players:

```
Lock-On Greece / DCS World Greece - EWRS v11.8
```

In dcs.log, search for EWRS: to see initialization messages, radar counts, and per-player menu events.

2.3 No Mission Editor Zones Required

Unlike some other systems, EWRS requires no trigger zones in the Mission Editor. It works entirely from the in-game radar detection engine and the coalition sensor network. Simply load the script and configure the Config.EWRS block at the top of the file.

3. Configuration Reference

All settings are in the Config.EWRS table at the top of the script. This is the only section you need to edit.

3.1 Core Parameters

Parameter	Default	Description
enable	true	Master switch. Set false to disable EWRS entirely.
refresh_time	45 s	How often the auto-broadcast report fires for each subscribed player.
message_display_duration	20 s	How long each report message stays on screen.
radar_cache_refresh_time	180 s	How often the radar unit cache is rebuilt (picks up dynamically spawned units).
default_max_contacts	6	Maximum enemy contacts shown per report (per player).
default_radius_nm	100 NM	Default enemy detection radius when a player first subscribes.
auto_enable_on_spawn	true	Automatically subscribe players to EWRS when they spawn into a slot.
friendly_picture_enable	true	Enable the Friendly Picture feature in the F10 menu.
default_friendly_max_contacts	4	Maximum friendly contacts shown per Friendly Picture report.
friendly_default_radius_nm	30 NM	Default radius for Friendly Picture.
enable_training_mode	true	Allow players to toggle Training Mode from the F10 menu.
enable_speed_reporting	false	Include airspeed in contact reports. Disabled by default.
speed_report_units_blue	KNOTS	Speed units for Blue coalition (KNOTS or KMH).
speed_report_units_red	KMH	Speed units for Red coalition.
default_units_blue	IMPERIAL	Blue coalition: IMPERIAL (NM/ft) or METRIC (km/m).
default_units_red	METRIC	Red coalition: IMPERIAL or METRIC.
default_reference_blue	BULLSEYE	Blue coalition bearing reference: BULLSEYE or OWN (ownship).
default_reference_red	OWN	Red coalition bearing reference.
only_armed_ships	true	When Ships radar is enabled, restrict to armed/heavy ships and carriers only.

3.2 Radar Source Types

The radar_types table controls which DCS unit attribute categories are used as radar sensors. Only units matching an enabled category contribute to the contact picture.

radar_types Key	Default	What It Covers
SAM_SR	true	SAM Search Radars (acquisition/search radar units)
SAM_TR	true	SAM Tracking Radars (fire control/guidance radars)
EWR	true	Dedicated Ground EWR units (e.g. 55G6, 1L13, P-18)
AWACS	true	Airborne Warning and Control System aircraft
AEW	true	Airborne Early Warning aircraft
Ships	true	Naval vessels (subject to only_armed_ships filter)
Fighter	false	Fighter aircraft — disabled by default to avoid false positives
Bomber	false	Bomber aircraft — disabled by default

ℹ *Setting Fighter or Bomber to true is not recommended in most missions — all aircraft of that type would act as sensors, which can create unrealistic contact reports.*

3.3 Range and Contact Limit Options

These tables define which values appear in the F10 Settings menu. Players can only select from the values listed here. Add or remove entries to suit your mission.

Option Set	Available Values (NM)	Default
enemy_range_options	30, 60, 80, 120, 180	100 NM (on spawn)
friendly_range_options	30, 60, 80, 120, 160	30 NM (on spawn)
enemy_contact_options	2, 4, 6, 8	6 contacts
friendly_contact_options	2, 4, 6, 8	4 contacts

4. Player Guide

4.1 Getting Started

When you spawn into any player slot, EWRS will automatically add a radio menu entry under F10 Other. If auto_enable_on_spawn is true (the default), you will also be subscribed to automatic GCI reports immediately. You will see a brief on-screen confirmation message showing your default enemy and friendly detection radii.

4.2 F10 Menu Structure

All EWRS controls are accessed through the F10 → Other → EWRS – [Your Name] menu tree:

Menu Path	Function
EWRS – [Player Name]	Root menu entry, labelled with your pilot name
→ Toggle HELO	Include or exclude helicopters from enemy reports
→ Bogey Dope	Immediate on-demand enemy picture (does not wait for refresh)
→ Request Friendly Picture	On-demand friendly air picture within your friendly radius
→ Toggle Training Mode	Switch between normal GCI mode and Training Mode
→ Settings	Sub-menu for all adjustable parameters
→ Set Enemy Range Filter → [NM]	Choose your enemy detection radius
→ Set Friendly Range Filter → [NM]	Choose your friendly detection radius
→ Set Reference → Ownship / Bullseye	Change bearing reference point
→ Set Units → Imperial / Metric	Switch between NM/ft and km/m
→ Set Enemy Max Contacts → [2/4/6/8]	Cap the number of enemy contacts displayed
→ Set Friendly Max Contacts → [2/4/6/8]	Cap the number of friendly contacts displayed
→ Speed Reporting → Toggle / Knots / km/h	Enable/disable and set speed units
→ Enable Reports	Subscribe to auto-refresh reports
→ Hide Reports	Unsubscribe from auto-refresh reports

4.3 Auto-Refresh Reports

When subscribed (via Enable Reports or auto-spawn), EWRS sends you an updated enemy air picture every 45 seconds (configurable). The report appears as on-screen text and stays visible for 20 seconds. If there are no contacts within your range filter, no message appears.

4.4 Bogey Dope

Select Bogey Dope from the root menu to request an immediate enemy picture. This is independent of the auto-refresh timer — useful when you need a snapshot right now before committing to an intercept. If you are in Training Mode, this will show your Training Mode picture instead.

5. Understanding the Reports

5.1 Enemy Picture Format

Each enemy contact line in the report follows this format (fields vary by unit/metric setting and whether speed reporting is enabled):

```
TYPE     BRG / RNG / ALT / ASPECT
TYPE     BRG / RNG / ALT / SPEED / ASPECT   (speed reporting on)
```

Field	Imperial Example	Metric Example
Unit type	MiG-29S	MiG-29S
Bearing	045 (degrees magnetic)	045
Range	62 nm	115 km
Altitude	25K (25,000 ft)	7K m (7,000 m)
Speed (if enabled)	450 kts	835 km/h
Aspect	HIGH ASPECT / MEDIUM ASPECT / LOW ASPECT	same

5.2 Aspect Explained

Aspect describes the target's flight path relative to your position, calculated from the target's tail:

Aspect Term	Angle off Tail	Meaning
LOW ASPECT	0° – 45°	Target is flying away from you (cold)
MEDIUM ASPECT	46° – 90°	Target is flying across your line of sight
HIGH ASPECT	91° – 180°	Target is flying towards you (hot)

ℹ *HIGH ASPECT means the contact is pointing towards you — potential intercept geometry. LOW ASPECT means it is flying away — likely egressing.*

5.3 Bearing Reference

Contacts can be reported in two reference frames, switchable per player in Settings:

- Ownship (OWN): Bearing is measured from your aircraft's current position. Intuitive for direct intercept use.
- Bullseye: Bearing and range are from the coalition bullseye point (set in the Mission Editor). Used for standardised GCI callouts across multiple players.

5.4 Altitude Display

Altitude is shown rounded to the nearest thousand feet (imperial) or thousand metres (metric), displayed as e.g. 25K (25,000 ft) or 7K m. Below 1,000 ft/m the raw figure is shown.

5.5 Contact Priority

Contacts are sorted by range (closest first). If you have set a max contacts limit below the actual number of detections, only the closest contacts are displayed. Adjust via Settings → Set Enemy Max Contacts.

6. Friendly Picture

Select Request Friendly Picture from the F10 menu to see a snapshot of friendly aircraft (both player-crewed and AI) within your friendly radius. This scans all coalition aircraft directly — it does not rely on radar detection — so all flying friendlies will appear regardless of radar coverage.

6.1 Report Format

```
FRIENDLY PICTURE - [Name]: 30nm - 3 contacts

Callsign    045 /  22 nm / 18K
F-16C       120 /  15 nm / 5K
```

If friendly_use_player_names is true (the default), player-occupied slots are labelled with the player's name. AI aircraft are labelled with their DCS unit type name.

6.2 Adjusting the Friendly Radius

Go to Settings → Set Friendly Range Filter and select from the available options (30, 60, 80, 120, or 160 NM). The default is 30 NM.

7. Training Mode

Training Mode completely changes what EWRS reports to you. Instead of the coalition air picture, it shows which enemy radar units are currently detecting (tracking) your own aircraft, along with their type, bearing from you, and range.

7.1 When to Use Training Mode

- SEAD/DEAD practice — learn which SAM radars are illuminating you before committing to a suppression run.
- Low-level ingress training — identify the moment you enter radar coverage.
- Electronic warfare training — assess the threat picture before employing jamming or chaff.

7.2 Training Mode Report Format

```
TRAINING MODE – [Name] – IMPERIAL:

SA-10 S-300PS SR 64H6E   045 /  42 nm
SA-6 Kub 1S91            210 /  18 nm
```

Contacts are sorted by range, closest first. The unit type shown is the exact DCS type name of the tracking radar, not a generic category.

7.3 Toggling Training Mode

Select Toggle Training Mode from your F10 EWRS menu. An on-screen confirmation message will confirm it is ENABLED or DISABLED. Both Bogey Dope and the auto-refresh report respect your current mode.

⚠ Training Mode and the normal enemy picture are mutually exclusive. While Training Mode is active, you will not receive standard GCI contact reports.

8. Per-Player Settings Reference

All settings are per-player and persist until you leave the slot or the mission ends. They are reset to the configured defaults when you respawn.

Setting	Options	Notes
Enemy Range Filter	30 / 60 / 80 / 120 / 180 NM	Only contacts within this radius appear in your reports
Friendly Range Filter	30 / 60 / 80 / 120 / 160 NM	Only friendlies within this radius appear in Friendly Picture
Enemy Max Contacts	2 / 4 / 6 / 8	Cap on how many enemy contacts are displayed, closest first
Friendly Max Contacts	2 / 4 / 6 / 8	Cap on how many friendly contacts are displayed
Reference	Ownship / Bullseye	Bearing reference for enemy contacts
Units	Imperial / Metric	Switches between NM + ft and km + m for all reports
Toggle HELO	On / Off (toggle)	Include or exclude helicopters from the enemy picture
Speed Reporting	On / Off + Knots / km/h	Adds airspeed to each contact line (global toggle, affects all players)
Enable / Hide Reports	Subscribe / Unsubscribe	Control whether you receive the auto-refresh broadcast
Training Mode	On / Off (toggle)	Switch between GCI mode and radar-detection training mode

ℹ *Speed Reporting is a global config toggle — changing it via the F10 menu affects all players on your coalition, not just yourself.*

9. How It Works (Technical Summary)

9.1 Radar Cache

On startup (and every radar_cache_refresh_time seconds thereafter) EWRS scans all DCS groups across all coalitions and builds a cache of units that match the enabled radar_types. This cache is used as the set of sensors for contact detection. The periodic refresh ensures dynamically spawned radar units are picked up without restarting the script.

9.2 Contact Detection

Every refresh_time seconds, EWRS iterates the radar cache and polls each active radar's DCS controller for its detected targets. Contacts from all radars are deduplicated by unit name, then filtered to enemy coalition only, and stored in each coalition's picture table. This picture table is also what IADS v5 reads when making SAM zone decisions.

9.3 Filtering for Display

When generating a report for a player, the picture is filtered further: ground radar units (SAM SR, SAM TR, EWR, Ships) are excluded from the air picture unless they are also classified as airborne (AWACS/AEW). Helicopters are included or excluded based on the player's current setting. Contacts are then filtered by the player's range radius and capped at their max contacts setting.

9.4 Two Coalition Instances

EWRS creates two independent EWRS objects at startup — one for Blue, one for Red. Each polls only its own coalition's radars and builds a picture of the opposing coalition's aircraft. Players receive reports only from their own coalition's EWRS instance.

9.5 Menu Lifecycle

The F10 menu entry is added 1 second after a player spawns (BIRTH event) and removed on death, ejection, crash, or slot change. If a player leaves a slot and re-slots, the menu is rebuilt fresh from defaults. This prevents stale menu entries accumulating in long multiplayer sessions.

10. Troubleshooting

Symptom	Cause / Fix
No EWRS menu in F10	MIST not loaded before EWRS. Ensure MIST.lua fires first in your ONCE trigger chain.
Menu appears but no reports arrive	auto_enable_on_spawn may be false. Use Enable Reports in the Settings menu, or set it true in config.
Reports show NO DATA	No radars detected contacts. Verify radar_types config and that friendly radar units are alive and active.
Contacts appear from wrong coalition	The EWRS is per-coalition. Each side only sees enemies. Check you are in the correct slot.
Ground units appear in reports	SAM SR/TR showing if they are also airborne — unlikely. Check isAirborneRadar logic is not filtering them out incorrectly.
Friendly Picture shows no contacts	All friendlies may be outside your friendly range filter. Increase it via Settings → Set Friendly Range Filter.
Training Mode shows no threats	No enemy radars are currently detecting you. Move into a SAM engagement envelope.
Menu duplicated / missing after slot change	Normal behaviour — menu is rebuilt on each BIRTH event. Re-slot and it will reappear.
Speed not shown in reports	enable_speed_reporting is false by default. Toggle via Settings → Speed Reporting.
Dynamically spawned radars not tracked	Wait up to 180 s for radar_cache_refresh_time to rebuild the cache.

11. Glossary

Term	Definition
EWRS	Early Warning Radar System — this script. Aggregates radar contacts and delivers them as GCI-style reports.
MIST	Mission Scripting Tools — a DCS Lua library required by EWRS for unit utilities and math helpers.
GCI	Ground Controlled Intercept — a real-world role where ground radar operators guide pilots onto targets.
Picture	The current set of detected enemy (or friendly) aircraft tracked by EWRS.
Bogey Dope	An on-demand request for an immediate contact report, without waiting for the auto-refresh.
Training Mode	An EWRS mode that shows which enemy radars are tracking the player rather than the general air picture.
Bullseye	A coalition reference point set in the Mission Editor. Used as an alternative bearing/range origin.
Ownship (OWN)	Bearing and range calculated relative to the player's current aircraft position.
Aspect	The target's flight path angle relative to the line between you and the target. HIGH = hot, LOW = cold.
Radar Cache	The pre-built list of radar-capable units used as sensors. Refreshed periodically to capture late spawns.
SAM SR / TR	SAM Search Radar / Tracking Radar — DCS unit attribute categories used to classify ground radar types.
EWR	Dedicated Early Warning Radar — typically long-range search radars like the 55G6 or P-18.
AWACS / AEW	Airborne Warning and Control System / Airborne Early Warning — radar-equipped aircraft used as sensors.
auto_enable_on_spawn	Config flag that automatically subscribes a player to EWRS reports the moment they spawn.